# Quasi-Random Photon Launching in MoCHII: Implementation, Validation, and the Diffuse-Bin Fix

**Abstract**

MoCHII gains an optional randomized quasi-Monte Carlo (RQMC) photon launch: Owen-scrambled Sobol coordinates replace the Mersenne Twister draws used to select a packet's source component, frequency bin, direction, and entry surface, while every post-launch transport event (forced interaction, Henyey–Greenstein scattering, Russian roulette) stays pseudo-random. The Sobol point is addressed by the *global* photon number, so the launch set is independent of the MPI task count. On the fixed-state analytic attenuation gate the cell-wise $\Gamma_{\rm HI}$ error scales as $\sim 1/N$ rather than the Monte Carlo $1/\sqrt{N}$, giving effective photon gains of $29\times$ at $2^{17}$ packets and $222\times$ at $2^{20}$, at unchanged wall time and with no significant bias. The stage-2 source gates (multiple point sources, external field) improve in step, and a diffuse-field variant reduces the replicate scatter of the ionized volume by $1.73\times$. This work also uncovered and fixed an unrelated defect in the diffuse-field energy-to-bin mapping that had become active once threshold-aligned bins (`par%ion_align_edges`) became the default: a closed-form log-grid formula mis-binned recombination photons emitted at an ionization threshold, sending He I ground-recombination photons into the bin below the He I edge; the corrected binary-search mapping lowers the mean neutral-helium fraction on a diffuse Strömgren sphere by 50.5%. The design rationale is recorded in `docs/QUASI_RANDOM_LAUNCH.md`.

## Contents

# 1. Motivation and design

MoCHII's ionizing band has no scattering: given a packet's launch frequency, position, and direction, the attenuation along its ray is integrated *analytically* (the zero-variance edge walk). Consequently the residual noise in the stellar radiation field comes almost entirely from the sampling of the launch variables — which source component, which frequency bin, and which direction each packet takes — and *not* from the transport estimator. This is exactly the situation an RQMC launch is built for: a low-dimensional integral with a deterministic post-launch response, so a low-discrepancy launch set directly reduces the noise in global photoionization rates, ionized volume, and the iteration-to-iteration convergence measures without touching the transport.

The design is deliberately hybrid. Only the fixed-dimensional launch step uses scrambled Sobol coordinates; every conditional, variable-length draw after launch stays on the Mersenne Twister. Dust-scattering histories consume a variable number of uniforms whose meaning depends on the packet's previous events, so mapping them onto sequential Sobol dimensions would be both unstable and high-dimensional. Keeping them pseudo-random preserves the unbiased scattered field and confines the low-discrepancy structure to the launch cube, where it helps most.

Two properties of MoCHII make the launch cube favorable. First, the frequency variable is a *discrete bin index*: inverting a stratified Sobol coordinate through the band luminosity CDF automatically delivers near-proportional packet counts in every bin (for $N = 2^m$ a bin of probability $p$ deviates from $pN$ by order one), removing the binomial bin-count noise that plain Monte Carlo inherits. Second, the launch is at most seven-dimensional (nine for the diffuse field), well within the range where scrambled Sobol nets retain useful variance reduction even though AMR cell boundaries and ionization fronts make the integrand only piecewise smooth. No ideal $N^{-1}(\log N)^d$ rate is assumed; the gain is measured (Section 5.).

# 2. Method

## 2.1 Sobol point by global index

The generator (`src/qmc_mod.f90`) is self-contained; no external library is used. The unscrambled Sobol integer for the 0-based global photon index $i$ and dimension $d$ is the random-access XOR of direction numbers over the set bits of $i$,

$$x_d(i) = \bigoplus_{k\,:\,\mathrm{bit}\,k\,\mathrm{of}\,i\,\mathrm{set}} V_d(k), \tag{1}$$

with 32-bit direction numbers $V_d(k)$. The cost is $O(\mathrm{popcount}(i))$ integer XORs, with no stored state and no sequential dependence, so any packet's coordinates are computed directly from its global number. Dimension 1 is the van der Corput sequence in base 2 ($V_1(k) = 2^{32-k}$); dimensions 2–12 use the Joe & Kuo (2008) `new-joe-kuo-6` primitive polynomials and initial direction integers, expanded by the Joe–Kuo recurrence. `QMC_MAXDIM` = 12 covers the seven-dimensional stellar superset and the nine-dimensional diffuse launch. The unscrambled coordinates reproduce `scipy.stats.qmc.Sobol` (which uses the same table) exactly.

## 2.2 Owen scrambling by hashing

A bare Sobol sequence has no ordinary statistical error estimate and can form coherent patterns with a Cartesian or octree grid, so the coordinates are Owen-scrambled. Full tree-based scrambling is unnecessary; MoCHII uses hash-based nested-uniform scrambling (Laine & Karras 2011; Burley 2020): reverse the 32 bits of $x_d$, apply a keyed Laine–Karras avalanche permutation (each bit is allowed to affect only less-significant bits), and reverse back. This is $O(1)$ per coordinate and preserves the net's equidistribution while producing statistically independent randomized replicates. The scrambled integer $k$ maps to the open unit interval as

$$u = (k+\tfrac{1}{2})2^{-32} \in (0,1), \tag{2}$$

so every coordinate is strictly interior, removing a failure class for any later mapping that takes a logarithm.

Two decorrelated scramble streams share the one Sobol net: the stellar launch (`QMC_STREAM_STELLAR`) and the diffuse launch (`QMC_STREAM_DIFFUSE`). A distinct 32-bit key for each dimension and stream is derived deterministically from `par%qmc_seed` by an integer avalanche hash, so the two streams behave as independent replicates while both remain balanced scrambles of the same net.

## 2.3 Fixed dimension layouts

The meaning of every dimension is fixed and never reassigned conditionally: a component-dependent reassignment would shift all later coordinates and destroy the intended low-dimensional projections. A point component simply leaves its surface dimensions unused; unused-but-fixed dimensions are harmless because the projections onto the consumed subset remain those of the scrambled net.

A single internal point source (stage 1) uses the three-dimensional layout of Table 1; every other source configuration uses the seven-dimension superset of Table 2; diffuse recombination packets use the nine-dimension layout of Table 3 on the separate diffuse stream.

Table 1: Single internal point source (`QMC_NDIM_USED` = 3).

| dim | variable | mapping |
|---|---|---|
| 1 | frequency bin | discrete band luminosity CDF |
| 2 | polar direction | $\mu = 2u - 1$ |
| 3 | azimuth | $\phi = 2\pi u$ |

## 2.4 Common random numbers and replicates

Using the same scrambled launch set at every ionization/thermal iteration is a form of common random numbers: it correlates the radiation estimates before and after an opacity update and reduces noise in their difference, which stabilizes

Table 2: Superset layout for every non-single-point configuration (multiple point sources, external field, or a mixed set).

| dim | variable | mapping |
|---|---|---|
| 1 | source component | component luminosity CDF |
| 2 | frequency bin | that component's spectrum CDF |
| 3 | polar / incidence | point: $\mu = 2u-1$; external: Lambert $\mu = \sqrt{u}$ |
| 4 | azimuth | $\phi = 2\pi u$ |
| 5 | entry surface | rectangular entry face / sphere entry $\cos\theta$ |
| 6 | first surface coordinate | in-face coordinate / sphere entry azimuth |
| 7 | second surface coordinate | in-face coordinate (unused for the sphere) |

Table 3: Diffuse recombination packets (`par%diffuse_field`), on the diffuse stream.

| dim | variable | mapping |
|---|---|---|
| 1 | emitting leaf | leaf luminosity CDF (binary search) |
| 2–4 | position in leaf | uniform, $2u-1$ per axis |
| 5 | polar direction | $\mu = 2u-1$ |
| 6 | azimuth | $\phi = 2\pi u$ |
| 7 | emission channel | channel CDF (H II/He II/He III/He I) |
| 8 | first energy variate | Exp(1) for the ground channels; He I branch pick |
| 9 | second energy variate | He I two-photon flat energy (unused otherwise) |

the convergence measures and separates front motion from sampling noise. The scramble key is `par%qmc_seed`, held fixed through all nonlinear iterations of one solve. A fixed scramble can, however, imprint a repeatable quadrature pattern that the nonlinear solution adapts to, making iteration differences look quieter than the true sampling uncertainty; convergence under one scramble is therefore not a substitute for replicate agreement. For a production error estimate, run independent replicates with different `qmc_seed` values — at least four, more when cost permits — and take their scatter as the RQMC uncertainty.

# 3. Implementation

The generator lives in `src/qmc_mod.f90`; the launch mappings that consume its coordinates are `gen_ion_photon_qmc` in `src/ion_band_mod.f90` (stellar) and `gen_diffuse_photon_qmc` in `src/diffuse_mod.f90` (diffuse). Two parameters control the option:

- `par%launch_sequence = 'random'` (default; the Mersenne Twister MT19937-64 path, arithmetically unchanged) or `'sobol'`;
- `par%qmc_seed` (default `12345`) — the scramble/replicate key, fixed across the gas iterations.

The stellar loop in `src/main.f90` addresses the Sobol point by the global photon number $i-1$ (0-based), and the diffuse loop does the same on its own stream indexed by the global diffuse photon number. Because the coordinates depend only on the global index and not on which rank handles a packet, the launch *set* is identical for any MPI task count; static or strided work assignment does not change the sample.

Everything after launch stays pseudo-random: forced first interaction, Henyey–Greenstein direction sampling, subsequent optical depths, and Russian roulette all draw from the rank-local Mersenne Twister exactly as before. Bitwise MPI-count independence of a *full* run is therefore obtained only in absorption-only transport, where the post-launch path is deterministic; with dust scattering enabled the launch set is still reproducible but the scattered histories are not. The multi-wavelength dust/SED launch (`src/sed_mod.f90`) is not exercised by the ionizing workflow and is left on its alias sampler; its inverse-CDF conversion is documented as unnecessary for now.

# 4. The diffuse bin-mapping fix

This work uncovered a defect independent of QMC, in the diffuse-field packet generator, and fixes it.

**The bug.** `gen_diffuse_photon` mapped a sampled photon energy to its band bin with a closed-form expression that assumed the legacy log-uniform bin grid. Once threshold-aligned bins (`par%ion_align_edges`) became the default, the bin edges sit *on* the ionization thresholds, and the closed form mis-assigns an energy at a threshold: a 24.587 eV He I ground-recombination photon landed in the bin *below* the He I edge, where it could never ionize helium.

**The fix.** The band edge array is now stored once (`ion_eedge`, valid for both grids), and `ion_bin_of(eph)` finds the bin by binary search: `ion_eedge`($b$) $\leq E_{\mathrm{ph}} <$ `ion_eedge`($b$+1), so an energy exactly on a threshold edge falls in the bin *above* it — the physically correct side for a recombination photon emitted at threshold $+\,\delta$. Both the pseudo-random and the QMC diffuse generators call `ion_bin_of`.

**Measured impact.** On a converged diffuse-field Strömgren sphere on the aligned (default) grid, buggy versus fixed: the mean $x_{\mathrm{HeI}}$ drops 50.5% (0.0147 $\rightarrow$ 0.0073), the mean $x_{\mathrm{HeII}}$ rises 0.75%, and $V_{\mathrm{ion}}/R_{\mathrm{eff}}$ are unchanged (hydrogen is stellar-dominated). Unit checks: on the legacy log grid the binary search reproduces the closed form over a two-million-point energy scan except at a single exact-edge floating-point tie; on the aligned grid the H I/He I/He II thresholds are bin edges and threshold $+\,\delta$ lands in the bin above. Runs with `diffuse_field` off are arithmetically untouched, and the recorded diffuse validations in the project log predate the aligned-edge default, so the bug had contaminated no recorded gate.

# 5. Validation

Every number below is measured and independently verified.

## 5.1 Generator gates (Q0/Q2)

The unscrambled Sobol coordinates for dimensions 1–12 match `scipy.stats.qmc.Sobol(d=12, scramble=False)` exactly (sorted integer lattices, maximum code difference 0). The Owen-scrambled sets are strictly interior; the onedimensional dyadic counts are exact at every level; the dimension-$(1,2)$ $(0,m,2)$-net and the elementary-interval occupancy are preserved under scrambling, for seeds `12345/777/2024` and both streams. Different seeds give different but individually balanced nets (maximum coordinate shift $|\Delta u| = 0.84$–$0.98$), and the stellar and diffuse streams differ like independent replicates.

## 5.2 Default-path regression

With `launch_sequence='random'`, at one MPI task under `-fp-model precise`, the pre-change and post-change builds are *bit-identical* for single-source, multi-source, and external inputs (worst $|\Delta| = 0$). The diffuse log-grid input sits at the code-generation rounding floor (maximum relative difference $6.8 \times 10^{-16}$, with the bins proven identical). At more than one task the difference equals the MPI `ALLREDUCE` reordering floor ($\sim 1$ ULP, the same as running one binary twice).

## 5.3 MPI-count independence

The Sobol launch set is task-count independent: `'sobol'` rates at 4 tasks versus 8 tasks agree to $6.0 \times 10^{-15}$ (stage 1) and $2.5 \times 10^{-14}$ (stage 2) relative, whereas the `'random'` contrast differs at order unity (each rank owns a different Mersenne Twister stream).

## 5.4 Variance gate: analytic attenuation

The cleanest test is the fixed-state G0 gate (`tests/g0_gamma`): $\Gamma_{\rm HI}$ against the exact analytic attenuation, comparing four ordinary Monte Carlo replicates with four RQMC replicates. Table 4 reports the cell-wise and radial-profile RMS error and the effective photon-number gain (the number of Monte Carlo packets needed to reach the RQMC RMS).

Table 4: Fixed-state $\Gamma_{\rm HI}$ error, Monte Carlo (MC) versus RQMC, four replicates each. "Gain" is the effective photon-number gain.

| | MC RMS | RQMC RMS | gain |
|---|---|---|---|
| $N = 2^{17}$ *packets* | | | |
| cell-wise | 16.9% | 3.1% | 28.8× |
| radial profile | 0.58% | 0.13% | 20.4× |
| $N = 2^{20}$ *packets* | | | |
| cell-wise | 6.0% | 0.40% | 222× |
| radial profile | 0.24% | 0.12% | 4.1× |

The RQMC cell-wise error scales as $\sim 1/N$ (Monte Carlo $\sim 1/\sqrt{N}$), so the gain grows with $N$. The RQMC bias is $\sim 0.06\%$, stable, and smaller than the Monte Carlo bias (0.12–0.41%): no significant bias. One caveat is reported honestly: at $2^{20}$ the RQMC radial-profile error saturates at a systematic floor (all four replicates give an identical 0.119%, with a fixed −0.06% bias) that is band-discretization systematics, not sampling noise, so the 4.1× profile gain there is a lower bound. Sequence generation is a few tens of integer operations per packet, negligible against ray traversal, so the wall time is unchanged.

### 5.5 Full-physics Strömgren smoke

On a full-physics Strömgren run, $R_{\rm eff}$ under `'sobol'` differs from `'random'` by $-0.039\%$ (within noise). The launch-noise reduction shows up directly in the convergence measure: the `'sobol'` run reached the volume-integrated criterion in 57 iterations where the `'random'` run hit the 60-iteration cap unconverged. The $x_{\rm HI}$ slices are clean — no angular spokes and no grid-aligned artifacts (inspected).

### 5.6 Stage-2 source gates

The existing stage-2 analytic checkers, rerun under `'sobol'` at their recorded criteria, improve or match the `'random'` numbers (Table 5).

Table 5: Stage-2 source gates under `'sobol'` versus the recorded `'random'` values.

| gate | `'sobol'` | `'random'` |
|---|---|---|
| two-point superposition (median) | 0.14% | 0.76% |
| two-point superposition (mean ratio) | 0.9989 | — |
| external rec, interior $\langle J\rangle/J$ | 0.9984 | 0.9981 |
| external sph, interior $\langle J\rangle/J$ | 0.9990 | 0.9991 |
| mixed point+external (median) | 0.13% | 0.75% |

### 5.7 Diffuse variance

On a diffuse-field Strömgren sphere from a neutral start, the replicate scatter (standard deviation over three seeds) of $V_{\rm ion}$ and $R_{\rm eff}$ is $1.73\times$ smaller for RQMC than for Monte Carlo, and unbiased. This is a smaller gain than the stellar launch, as the design anticipated: the emitting-leaf CDF is strongly discontinuous and changes as the gas state iterates, and the diffuse cube is higher-dimensional.

## 6. Usage guidance and limitations

Set `par%launch_sequence = 'sobol'` to enable the RQMC launch and prefer a power-of-two photon count, `nphotons` $= 2^m$, where the Sobol net has its cleanest balance. Keep `par%qmc_seed` fixed across the nonlinear iterations of one solve (common random numbers), and change it only to generate independent replicates; take the scatter of at least four replicates as the RQMC error estimate. The default remains `'random'` so recorded gates reproduce.

The gains follow the design expectation. The largest reductions are for global, direct-field quantities: direct stellar photoionization rates, integrated ionized volume and effective Strömgren radius, volume-integrated convergence measures, smooth radial profiles, and early iterations run with a modest photon budget. Smaller or inconsistent gains are expected for the maximum error in a single I-front cell, narrow shadow boundaries and clump skins, highly localized line emissivities, high-resolution image pixels, scattering-dominated paths, and diffuse-field-dominated cells. Low discrepancy does not remove physical discretization errors — frequency binning, AMR resolution, atomic-data fits, and nonlinear stopping tolerances remain independent error sources — and no universal speedup factor should be assumed; a useful reduction is best reported as an effective photon-number gain. By design, variable-length scattering histories remain on the Mersenne Twister; extending RQMC into them would need a maximum-order dimension policy and a separate validation campaign.

## References


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